Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across four product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, discount usage and subscription behavior, so that the retail company can improve sales, customer satisfaction and long-term loyalty.

The work follows an end-to-end workflow: data preparation in **Python**, structured analysis in **MySQL**, an interactive dashboard in **Power BI**, and a set of business recommendations. The overarching business question is: *"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"*

Tools used: Python (pandas, SQLAlchemy, PyMySQL) in VS Code | MySQL | Power BI

2. Dataset Summary

- **Rows:** 3,900 **Columns:** 18
- **Customer demographics:** Age, Gender, Location, Subscription Status
- **Purchase details:** Item Purchased, Category, Purchase Amount (₹), Season, Size, Color
- **Shopping behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method
- **Missing data:** 37 values in the Review Rating column (all other columns complete)

Metric	Value
Customers / purchases	3,900
Average purchase amount	₹59.76
Purchase amount range	₹20 – ₹100
Average age	44.1 years (18 – 70)
Average review rating	3.75 (2.5 – 5.0)
Average previous purchases	25.4 (1 – 50)
Total revenue	₹233,081

3. Exploratory Data Analysis using Python

Data preparation and cleaning were carried out in a Jupyter notebook in VS Code:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `df.describe()` for summary statistics.
- **Missing Data Handling:** Found 37 null values in `Review Rating` and imputed them with the median rating of each product category.
- **Column Standardization:** Renamed columns to snake_case (for example `purchase_amount`) for readability and easier SQL use.
- **Feature Engineering:** Created **age_group** (Young Adult, Adult, Middle-aged, Senior) by splitting age into four quartile-based bins, and **purchase_frequency_days** by converting purchase frequency labels (Weekly = 7, Fortnightly = 14, Monthly = 30, Quarterly = 90, Annually = 365) into numbers of days.
- **Data Consistency Check:** Verified that `discount_applied` and `promo_code_used` were identical on every row, so the redundant `promo_code_used` column was dropped.
- **Database Integration:** Connected Python to MySQL using SQLAlchemy and PyMySQL, and loaded the cleaned DataFrame into the `customer` table of the `customer_behavior` database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer ten key business questions on customer segments, loyalty and purchase drivers.

1. Revenue by Gender

Compared total revenue generated by male vs. female customers.

Gender	Revenue (₹)
Female	75,191
Male	157,890

> ***Insight:*** *Male customers generate about 68% of total revenue, roughly double that of female customers (32%). 2,652 of the 3,900 customers are male.*

2. High-Spending Discount Users

Identified customers who used a discount but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
…	…
Total rows: 839	

> ***Insight:*** *839 purchases used a discount and still exceeded the average spend of ₹59.76, so discounts do not necessarily lower the basket size.*

3. Top 5 Products by Rating

Found products with the highest average review ratings.

Item purchased	Average product rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

> ***Insight:*** *Ratings are tightly clustered between 3.78 and 3.86, so no product stands out strongly. Gloves, Sandals and Boots lead and are good candidates for promotion.*

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

Shipping type	Avg purchase amount (₹)
Standard	58.46
Express	60.48

> ***Insight:*** *Express shippers spend about ₹2 (3.5%) more per purchase. The gap is small but suggests that paid fast delivery attracts slightly higher-value orders.*

5. Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

Subscription	Customers	Avg spend (₹)	Total revenue (₹)
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

> ***Insight:*** *Subscribers do* ***not*** *spend more: average spend is almost identical (₹59.49 vs. ₹59.87). Only 27% of customers subscribe, so non-subscribers drive 73% of revenue simply because there are more of them.*

6. Discount-Dependent Products

Identified the 5 products with the highest percentage of discounted purchases.

Item purchased	Discount rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

> ***Insight:*** *About half of the purchases of these items used a discount (43% across all purchases), so margins on Hats, Sneakers and Coats are the most exposed to discounting.*

7. Customer Segmentation

Classified customers into New (1 previous purchase), Returning (2–10) and Loyal (11+) segments based on purchase history.

Customer segment	Number of customers	Share
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%

> ***Insight:*** *Four in five customers are already Loyal, while New customers are only 2%. The base is strong, but there is little new-customer inflow to refresh it.*

8. Top 3 Products per Category

Listed the most purchased products within each category.

Rank	Category	Item purchased	Total orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

> ***Insight:*** *Jewelry, Blouse and Pants are the most purchased items (171 orders each). Order counts within each category are very close, so demand is spread evenly across products.*

9. Repeat Buyers & Subscriptions

Checked whether customers with more than 5 previous purchases are more likely to subscribe.

Subscription status	Repeat buyers
No	2,518
Yes	958

> ***Insight:*** *Of 3,476 repeat buyers, 958 (27.6%) subscribe, almost the same as the overall 27% rate. Repeat buying does not lead to subscribing on its own, which is an opportunity for targeted offers.*

10. Revenue by Age Group

Calculated the total revenue contribution of each age group.

Age group	Total revenue (₹)	Share
Young Adult	62,143	26.7%
Middle-aged	59,197	25.4%
Adult	55,978	24.0%
Senior	55,763	23.9%

> ***Insight:*** *Revenue is spread almost evenly across age groups. Young Adults contribute the most, but the gap to Seniors is under 3 percentage points.*

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** connected to the MySQL `customer` table to present the insights visually. Stakeholders can filter by subscription status, gender, category and shipping type.



Figure: Customer Behavior Dashboard in Power BI. This view is filtered to the “Next Day Air” shipping type (648 customers); with no filter applied it shows all 3,900 customers.

Dashboard highlights (all customers, no filters applied)

- 3.9K customers, an average purchase amount of ₹59.76 and an average review rating of 3.75.

- 27% of customers are subscribers; 73% are not.
- Clothing is the top category by revenue and sales volume, followed by Accessories, Footwear and Outerwear.
- Young Adult and Middle-aged groups lead on revenue and sales volume.

6. Key Findings

- **Gender gap:** male customers produce about 68% of revenue.
- **Subscriptions are under-used and not yet valuable:** only 27% subscribe, and subscribers spend no more than non-subscribers.
- **Strong loyalty base:** about 80% of customers are Loyal, but only 2% are New.
- **Heavy discounting:** 43% of purchases use a discount, and some products reach 50%.
- **Even demand:** revenue is balanced across age groups, and product ratings are all similar (3.78–3.86).

7. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers (free shipping, early access, member pricing). Today subscribers do not spend more, so the offer must add real value.
- **Customer Loyalty Programs** – Reward repeat buyers with points or tiers, and target repeat buyers who are not yet subscribers (2,518 customers).
- **Review Discount Policy** – Balance sales boosts with margin control, especially for Hats, Sneakers and Coats, where about half of the sales are discounted.
- **Product Positioning** – Highlight top-rated products (Gloves, Sandals, Boots) and best sellers (Jewelry, Blouse, Pants) in campaigns.
- **Targeted Marketing** – Focus efforts on Young Adult and Middle-aged customers and on Express-shipping users, and run campaigns to attract more female customers.
- **Acquire New Customers** – With only 83 New customers, invest in acquisition (referral offers, first-purchase incentives) to refresh the customer base.